



S3 PicoBot quick reference cards

Six one-page cards for the students' desk: Thonny · Python · MicroPython on the Pico · PicoBot library · pins and wiring · moves, roles and safety · Version 2

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How to use the cards

Print each card on one A4 page, laminate it, and keep one set on every team's table. The cards do not replace the lessons: they help you **remember** what you have already learned. Card 6 (team roles and safety) belongs on the table from the first session.

Card	Content	Needed from
1	Thonny quick guide	session 5
2	Python cheat sheet	session 5
3	MicroPython on the Pico: pins, time, PWM, ADC, I2C, Wi-Fi	session 8
4	The PicoBot library: motors, moves, arm	session 15
5	PicoBot pins and wiring	session 8
6	Mecanum moves, team roles and safety	session 1

Card 1 — Thonny quick guide

I want to ...	Do this
choose where the program runs	Tools → Options → Interpreter: <i>Local Python 3</i> (computer) or <i>MicroPython (Raspberry Pi Pico)</i> — or click the interpreter name at the bottom right
run the program	F5 or the green Run button
stop the program	Ctrl+F2 or the red Stop button; in the Shell Ctrl+C stops a running program
restart the Pico (soft reboot)	Ctrl+D in the Shell
save to the Pico	File → Save as... → Raspberry Pi Pico (name ends with <code>.py</code>)
make a program start by itself	save it on the Pico as main.py
see the files on the Pico	View → Files (top: this computer; bottom: Raspberry Pi Pico)
copy a file to the Pico	in View → Files , right-click the file on the computer → Upload to /
try one line	type it in the Shell after <code>>>></code> and press Enter
make the text bigger	View → Increase font size (Ctrl + +)
find an error	read the last line of the red message: error type + line number

Tip

Pico not found? Try another USB cable (some cables only charge). Check the interpreter at the bottom right. Press **Stop** once. More help: A5 Troubleshooting.

Card 2 — Python cheat sheet

Topic	Example	Note
Print	<code>print("Speed:", speed)</code>	several values, separated by commas
Variable	<code>speed = 50</code>	= stores a value
Types	<code>5 int · 7.4 float · "PicoBot" str · True bool</code>	<code>type(x)</code> shows the type
Input	<code>name = input("Name? ")</code>	always text; <code>int(input(...))</code> for a number
Maths	<code>+ - * / · //</code> whole division · <code>%</code> remainder · <code>**</code> power	<code>7 // 2 → 3</code> , <code>7 % 2 → 1</code>
Compare	<code>== != < > <= >=</code>	= stores, == compares
Logic	<code>and · or · not</code>	
If	<code>if d < 20: ... elif d < 50: ... else: ...</code>	colon + 4 spaces
While	<code>while battery > 7.0:</code>	repeats while True
For	<code>for i in range(4):</code>	<code>i = 0, 1, 2, 3</code>
Function	<code>def area(w, h): ... return w * h</code>	call: <code>area(3, 4)</code>
List	<code>runs = [21, 23, 22] · runs.append(20) · runs[0]</code>	first item has index 0
List tools	<code>len(runs) · sum(runs) · max(runs) · min(runs)</code>	
Dictionary	<code>robot = {"name": "Bolt", "speed": 50} · robot["speed"]</code>	key → value
Round	<code>round(22.456, 1) → 22.5</code>	
Module	<code>import time · from machine import Pin</code>	
Comment	<code># explains the code</code>	ignored by Python

Common errors

Error	Usually means
<code>SyntaxError</code>	writing rule broken: missing <code>:</code> or <code>)</code> , quotes not closed
<code>IndentationError</code>	spaces at the start of a line are wrong
<code>NameError</code>	name not defined yet, or spelled differently (<code>Speed ≠ speed</code>)
<code>TypeError</code>	wrong type, e.g. <code>"5" + 3</code>
<code>ImportError / ModuleNotFoundError</code>	the module file is not on the Pico, or the name is wrong

Card 3 — MicroPython on the Pico

Topic	Example
Onboard LED	<code>led = Pin("LED", Pin.OUT) · led.on() · led.off() · led.toggle()</code>
Digital input	<code>s = Pin(8, Pin.IN, Pin.PULL_UP) · s.value() → 0 or 1</code>
BOOTSEL button	<code>import rp2 · rp2.bootSEL_button() → 1 when pressed</code> (MicroPython 1.25 or newer)
Wait	<code>import time · time.sleep(0.5) · time.sleep_ms(20)</code>
Measure time	<code>start = time.ticks_ms() ... time.ticks_diff(time.ticks_ms(), start)</code>
Non-blocking loop	<code>if time.ticks_diff(time.ticks_ms(), last) >= 500: → do something, last = time.ticks_ms()</code>
Temperature (onboard)	<code>adc = ADC(4) · v = adc.read_u16() * 3.3 / 65535 · t = 27 - (v - 0.706) / 0.001721</code>
I2C	<code>i2c = I2C(1, sda=Pin(2), scl=Pin(3)) · i2c.scan() → [64] = 0x40</code>
Wi-Fi access point	<code>import network · ap = network.WLAN(network.AP_IF) · ap.config(essid="TeamName", password="12345678") · ap.active(True)</code>
Always stop	<code>try: ... finally: motors.StopAllMotors()</code>
Start automatically	save as <code>main.py</code> on the Pico

Note

The Pico's pins work with **3.3 V**. Never connect 5 V or the battery to a GP pin.

Card 4 — The PicoBot library

Files in the Pico's main folder: `picobot_motors.py`, `picobot_arm.py`, `pca9685.py` (from `picobot-setup`) and, for the PicoBot class, `picobot.py`.

One motor — MotorDriver

```
from picobot_motors import MotorDriver
motors = MotorDriver()
motors.TurnMotor("LeftFront", "forward", 50) # speed 0-100
motors.MotorStop("LeftFront")
motors.StopAllMotors()
```

Motors: "LeftFront" · "LeftBack" · "RightFront" · "RightBack" — directions: "forward" · "backward".

Moves — PicoBot (speed 0-100, default 50)

Move	Method	Move	Method
forward	<code>goForward()</code>	diagonal forward-right	<code>moveRightForward()</code>
backward	<code>goBackward()</code>	diagonal forward-left	<code>moveLeftForward()</code>
sideways right	<code>moveRight()</code>	diagonal backward-right	<code>moveRightBackward()</code>
sideways left	<code>moveLeft()</code>	diagonal backward-left	<code>moveLeftBackward()</code>
turn clockwise	<code>rotateRight()</code>	stop	<code>stopRobot()</code> / <code>hardStop()</code>
turn counter-clockwise	<code>rotateLeft()</code>	one wheel	<code>robot.m.TurnMotor(...)</code>

```
from picobot import PicoBot
robot = PicoBot() # the arm moves to 90 degrees!
robot.goForward(50)
```

Arm — PicoBotArm (channel 0 base, 1 arm, 2 gripper)

```
from picobot_arm import PicoBotArm
arm = PicoBotArm()
arm.control_servo(1, 120) # channel, angle
arm.smooth_move_servo(0, 45) # slowly
arm.reset_servos() # all back to 90
```

Servo	Channel	Safe range	Home
base	0	0-180°	90°
arm	1	40-140°	90°
gripper	2	40-140°	90°

Note

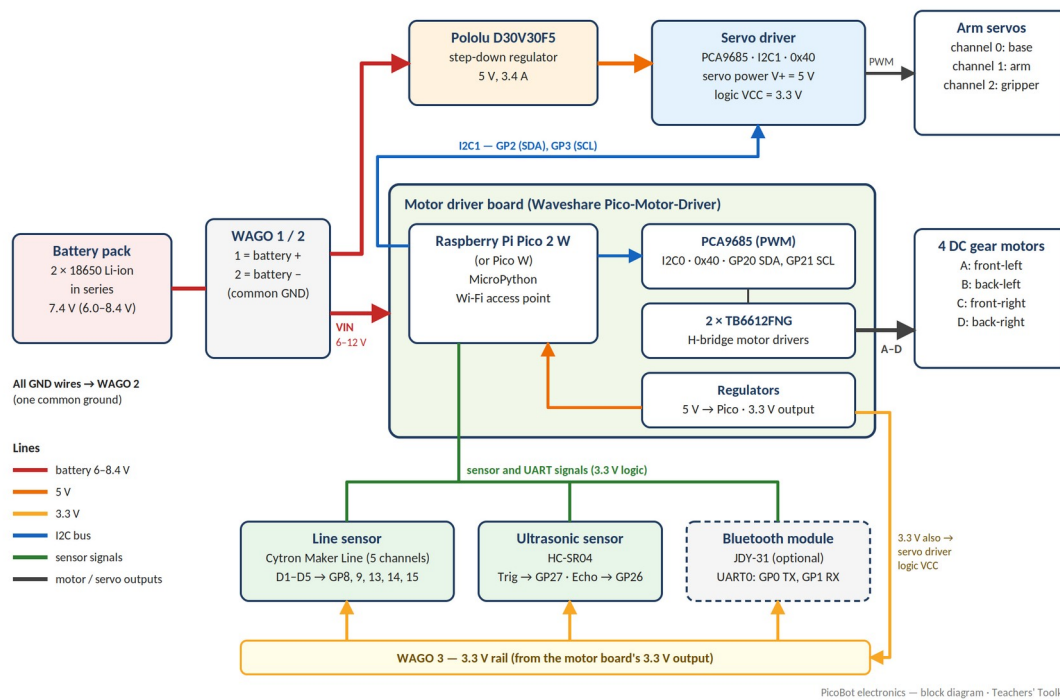
Stop before a diagonal move, and use the updated `picobot.py` (the old one waits 10 s in `stopRobot()`).

Card 5 — PicoBot pins and wiring

Part	Connection	Pico pin (GPIO)
Motor board PCA9685 (motors)	I2C0, address 0x40	GP20 SDA, GP21 SCL
Servo driver PCA9685 (arm)	I2C1, address 0x40	GP2 SDA, GP3 SCL
Line sensor D1 (right) ... D5 (left)	digital inputs, pull-up, 1 = line	GP8, GP9, GP13, GP14, GP15
Ultrasonic sensor HC-SR04	Trig / Echo	GP27 / GP26
Bluetooth JDY-31 (optional)	UART0: Pico TX → RXD, RX ← TXD	GP0 / GP1
Onboard LED	on the Wi-Fi chip	Pin ("LED")
Temperature sensor	onboard	ADC (4)

Power	Voltage
Battery pack: 2 × 18650 in series	7.4 V nominal, 8.4 V full — recharge at 7.0 V
WAGO 1 = battery +, WAGO 2 = common ground (GND)	6.0–8.4 V
Pololu regulator → servo power (V+)	5 V
WAGO 3 → sensors, servo-driver logic, JDY-31	3.3 V

Motor terminal	Wheel
A	front-left ("LeftFront")
B	back-left ("LeftBack")
C	front-right ("RightFront")
D	back-right ("RightBack")



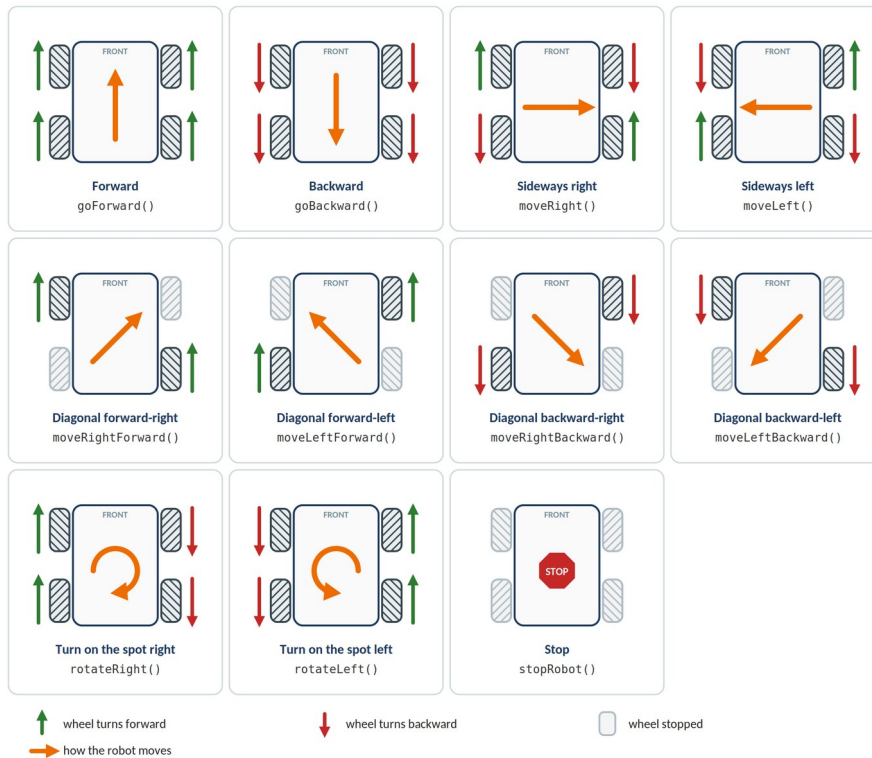
PicoBot electronics — block diagram · Teachers' Toolkit

PicoBot block diagram (A1).

Card 6 — Mecanum moves, team roles and safety

PicoBot — the mecanum moves of the library (picobot.py)

Seen from above, front at the top. Green = wheel forward, red = wheel backward, grey = wheel stopped.



PicoBot mecanum moves - Teachers' Toolkit - wheel directions as in picobot.py

Mecanum moves of the PicoBot library (seen from above).

Move	Front-left	Back-left	Front-right	Back-right
forward / backward	F / B	F / B	F / B	F / B
sideways right	F	B	B	F
sideways left	B	F	F	B
turn clockwise	F	F	B	B

Team roles — change after every task

Role	Job
Driver / programmer	drives with the web page or types and runs the code
Safety officer	switches on/off, checks the battery, lifts the robot for tests, calls "STOP!"
Engineer / tester	builds the set-up, measures, compares with the prediction
Documenter	writes predictions, results and the calibration record

Safety

- **Lift the robot** for the first test of every program that moves the wheels.
- Drive on the **floor**, never on a table. One hand ready.
- **Fingers away from the gripper** when the arm is powered.
- Battery **below 7.0 V** → **recharge**. Check + and - when inserting cells. Switch off after use.
- Something hot, smelling or smoking → **switch off**, take the cells out, tell the teacher.

Predict → test → measure → explain · Calibrate before you trust

About this toolkit



Co-funded by
the European Union



Teachers' ST(R)EM: Robots in STEM toolkit — the PicoBot Teachers' Toolkit, a result of Work Package 4 of the Erasmus+ project **ROBO STEAM ACADEMY** (KA220-VET — Cooperation partnerships in vocational education and training, project number KA220-VET-7CF4F308, 2024–2027). Project website: <https://robosteam.eu/>

Partners: National STEM Center (coordinator, Bulgaria, <https://stem.mon.bg/>) · Institute of Information and Communication Technologies, Bulgarian Academy of Sciences (Bulgaria, <https://www.iict.bas.bg/>) · Academy Nikola Tesla (Bulgaria, <https://ant.bg/>) · Technical University of Košice (Slovakia, <https://www.tuke.sk/>) · Stredná priemyselná škola elektrotechnická Prešov (Slovakia, <https://spseadlerka.edupage.org/>) · National Association of Public Librarians and Libraries in Romania – ANBPR (Romania, <https://anbpr.org.ro/>) · Liceul Tehnologic Cezar Nicolau (Romania, <https://www.cezar-nicolau.ro/>).

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